

Spring-data-jpa

Hands on 1

Write queries on country table using Query Methods

CountryRepository.java (additions)

```
package com.cognizant.ormlearn.repository;

import java.util.List;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;
import com.cognizant.ormlearn.model.Country;

@Repository
public interface CountryRepository extends JpaRepository<Country, String> {

    // 1. Search by containing text
    List<Country> findByNameContaining(String text);

    // 2. Same search, ascending order by name
    List<Country> findByNameContainingOrderByNameAsc(String text);

    // 3. Alphabet index - starts with given letter
    List<Country> findByNameStartingWith(String letter);
}
```

OrmLearnApplication.java (test methods)

```
private static void testSearchCountry() {
    LOGGER.info("Start");
    List<Country> countries = countryRepository.findByNameContaining("ou");
    LOGGER.debug("countries={}", countries);
    LOGGER.info("End");
}

private static void testSearchCountrySorted() {
    LOGGER.info("Start");
    List<Country> countries = countryRepository.findByNameContainingOrderByNameAsc("ou");
    LOGGER.debug("countries={}", countries);
    LOGGER.info("End");
}

private static void testCountryByAlphabet() {
    LOGGER.info("Start");
    List<Country> countries = countryRepository.findByNameStartingWith("Z");
    LOGGER.debug("countries={}", countries);
    LOGGER.info("End");
}
```

output — testSearchCountry() (unsorted, contains 'ou')

```
17-07-26 09:10:02.114 TRACE main org.hibernate.SQL select country0_.co_code as
co_code1_0_, country0_.co_name as co_name2_0_ from country country0_ where
country0_.co_name like ?
17-07-26 09:10:02.121 TRACE main o.h.type.descriptor.sql.BasicBinder binding
parameter [1] as [VARCHAR] - [%ou%]
17-07-26 09:10:02.129 DEBUG main OrmLearnApplication testSearchCountry countries=
[Country[BV,Bouvet Island], Country[DJ,Djibouti], Country[GP,Guadeloupe],
Country[GS, South Georgia and the South Sandwich Islands], Country[LU,Luxembourg],
Country[SS,South Sudan], Country[TF,French Southern Territories],
Country[UM,United States Minor Outlying Islands], Country[ZA,South Africa]]
```

output — testSearchCountrySorted() (ordered by name asc)

```
17-07-26 09:11:40.201 TRACE main org.hibernate.SQL select country0_.co_code as
co_code1_0_, country0_.co_name as co_name2_0_ from country country0_ where
country0_.co_name like ? order by country0_.co_name asc
17-07-26 09:11:40.208 DEBUG main OrmLearnApplication testSearchCountrySorted
countries=[Country[BV,Bouvet Island], Country[DJ,Djibouti], Country[TF,French
Southern Territories], Country[GP,Guadeloupe], Country[LU,Luxembourg],
Country[ZA,South Africa], Country[GS,South Georgia and the South Sandwich
Islands], Country[SS,South Sudan], Country[UM,United States Minor Outlying
Islands]]
```

output — testCountryByAlphabet() ('Z')

```
17-07-26 09:12:55.301 TRACE main org.hibernate.SQL select country0_.co_code as
co_code1_0_, country0_.co_name as co_name2_0_ from country country0_ where
country0_.co_name like ?
17-07-26 09:12:55.305 TRACE main o.h.type.descriptor.sql.BasicBinder binding
parameter [1] as [VARCHAR] - [Z%]
17-07-26 09:12:55.310 DEBUG main OrmLearnApplication testCountryByAlphabet
countries=[Country[ZM,Zambia], Country[ZW,Zimbabwe]]
```

Hands-on 2: Query Methods on stock table

Table creation

```
CREATE TABLE IF NOT EXISTS `ormlearn`.`stock` (
  `st_id` INT NOT NULL AUTO_INCREMENT,
  `st_code` varchar(10),
  `st_date` date,
  `st_open` numeric(10,2),
  `st_close` numeric(10,2),
  `st_volume` numeric,
  PRIMARY KEY (`st_id`)
);
```

Stock.java

```
package com.cognizant.ormlearn.model;
import java.math.BigDecimal;
import java.time.LocalDate;
import javax.persistence.*;
@Entity
@Table(name = "stock")
public class Stock {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "st_id")
    private int id;
    @Column(name = "st_code")
    private String code;
    @Column(name = "st_date")
    private LocalDate date;
    @Column(name = "st_open")
    private BigDecimal open;

    @Column(name = "st_close")
    private BigDecimal close;

    @Column(name = "st_volume")
    private long volume;
    // getters, setters
    @Override
    public String toString() {
        return "Stock [code=" + code + ", date=" + date + ", open=" + open
            + ", close=" + close + ", volume=" + volume + "]\n";
    }
}
```

OrmLearnApplication.java (test methods, StockRepository autowired directly)

```
private static void testFacebookSeptemberStocks() {
    LOGGER.info("Start");
    List<Stock> stocks = stockRepository.findByCodeAndDateBetween(
        "FB", LocalDate.of(2019, 9, 1), LocalDate.of(2019, 9, 30));
    LOGGER.debug("stocks={}", stocks);
    LOGGER.info("End");
}

private static void testGoogleHighStocks() {
    LOGGER.info("Start");
    List<Stock> stocks = stockRepository.findByCodeAndCloseGreaterThan("GOOGL", new BigDecimal("1250"));
    LOGGER.debug("stocks={}", stocks);
    LOGGER.info("End");
}

private static void testTopVolumeDates() {
    LOGGER.info("Start");
    List<Stock> stocks = stockRepository.findTop3ByOrderByVolumeDesc();
    LOGGER.debug("stocks={}", stocks);
    LOGGER.info("End");
}

private static void testNetflixLowestStocks() {
    LOGGER.info("Start");
    List<Stock> stocks = stockRepository.findTop3ByCodeOrderByCloseAsc("NFLX");
    LOGGER.debug("stocks={}", stocks);
    LOGGER.info("End");
}
```

output — testFacebookSeptemberStocks()

```
17-07-26 09:20:11.101 TRACE main org.hibernate.SQL select stock0_.st_id ... from
stock stock0_ where stock0_.st_code=? and stock0_.st_date between ? and ?
17-07-26 09:20:11.130 DEBUG main OrmLearnApplication testFacebookSeptemberStocks
stocks=[Stock[FB,2019-09-03,184.00,182.39,9779400], Stock[FB,2019-09-
04,184.65,187.14,11308000], ... Stock[FB,2019-09-27,180.49,177.10,14656200]] (19
rows)
```

Expected output — testGoogleHighStocks()

```
17-07-26 09:21:40.201 TRACE main org.hibernate.SQL select stock0_.st_id ... from
stock stock0_ where stock0_.st_code=? and stock0_.st_close>?
17-07-26 09:21:40.230 DEBUG main OrmLearnApplication testGoogleHighStocks stocks=
[Stock[G00GL,2019-04-22,1236.67,1253.76,954200], Stock[G00GL,2019-04-
23,1256.64,1270.59,1593400], Stock[G00GL,2019-04-24,1270.59,1260.05,1169800],
Stock[G00GL,2019-04-25,1270.30,1267.34,1567200], Stock[G00GL,2019-04-
26,1273.38,1277.42,1361400], Stock[G00GL,2019-04-29,1280.51,1296.20,3618400],
Stock[G00GL,2019-10-17,1251.40,1252.80,1047900]]
```

Expected output — testTopVolumeDates()

```
17-07-26 09:22:55.410 TRACE main org.hibernate.SQL select stock0_.st_id ... from
stock stock0_ order by stock0_.st_volume desc limit ?
17-07-26 09:22:55.418 DEBUG main OrmLearnApplication testTopVolumeDates stocks=
[Stock[FB,2019-01-31,165.60,166.69,77233600], Stock[FB,2018-10-
31,155.00,151.79,60101300], Stock[FB,2018-12-19,141.21,133.24,57404900]]
```

Expected output — testNetflixLowestStocks()

```
17-07-26 09:23:33.501 TRACE main org.hibernate.SQL select stock0_.st_id ... from
stock stock0_ where stock0_.st_code=? order by stock0_.st_close asc limit ?
17-07-26 09:23:33.508 DEBUG main OrmLearnApplication testNetflixLowestStocks
stocks=[Stock[NFLX,2018-12-24,242.00,233.88,9547600], Stock[NFLX,2018-12-
21,263.83,246.39,21397600], Stock[NFLX,2018-12-26,233.92,253.67,14402700]]
```

Hands-on 3: Payroll tables and bean mapping

Employee.java

```

package com.cognizant.ormlearn.model;

import java.util.Date;
import javax.persistence.*;

@Entity
@Table(name = "employee")
public class Employee {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "em_id")
    private int id;

    @Column(name = "em_name")
    private String name;

    @Column(name = "em_salary")
    private double salary;

    @Column(name = "em_permanent")
    private boolean permanent;

    @Column(name = "em_date_of_birth")
    private Date dateOfBirth;

    // getters, setters, toString()
}

```

Department.java

```

package com.cognizant.ormlearn.model;

import javax.persistence.*;

@Entity
@Table(name = "department")
public class Department {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "dp_id")
    private int id;

    @Column(name = "dp_name")
    private String name;

    // getters, setters, toString()
}

```

Skill.java

```

package com.cognizant.ormlearn.model;

import javax.persistence.*;

@Entity
@Table(name = "skill")
public class Skill {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "sk_id")
    private int id;

    @Column(name = "sk_name")
    private String name;

    // getters, setters, toString()
}

```

Repositories

```

public interface EmployeeRepository extends JpaRepository<Employee, Integer> {}
public interface DepartmentRepository extends JpaRepository<Department, Integer> {}
public interface SkillRepository extends JpaRepository<Skill, Integer> {}

```

Hands-on 4: @ManyToOne — Employee → Department

EmployeeService.java

```

@Service
public class EmployeeService {

    private static final Logger LOGGER = LoggerFactory.getLogger(EmployeeService.class);

    @Autowired
    private EmployeeRepository employeeRepository;

    @Transactional
    public Employee get(int id) {
        LOGGER.info("Start");
        return employeeRepository.findById(id).get();
    }

    @Transactional
    public void save(Employee employee) {
        LOGGER.info("Start");
        employeeRepository.save(employee);
        LOGGER.info("End");
    }
}

```

Employee.java (add relationship)

```

@ManyToOne
@JoinColumn(name = "em_dp_id")
private Department department;

// getter and setter for department

```

OrmLearnApplication.java

```

private static EmployeeService employeeService;
private static DepartmentService departmentService;
private static SkillService skillService;

// in main():
employeeService = context.getBean(EmployeeService.class);
departmentService = context.getBean(DepartmentService.class);
skillService = context.getBean(SkillService.class);

private static void testGetEmployee() {
    LOGGER.info("Start");
    Employee employee = employeeService.get(1);
    LOGGER.debug("Employee:{}", employee);
    LOGGER.debug("Department:{}", employee.getDepartment());
    LOGGER.info("End");
}

```

output — testGetEmployee()

```

17-07-26 09:30:12.101 INFO main EmployeeService get Start
17-07-26 09:30:12.108 TRACE main org.hibernate.SQL select employee0_.em_id as
em_id1_2_0_, employee0_.em_date_of_birth as em_date_2_2_0_, employee0_.em_dp_id as
em_dp_id6_2_0_, employee0_.em_name as em_name3_2_0_, employee0_.em_permanent as
em_perma4_2_0_, employee0_.em_salary as em_salar5_2_0_, department1_.dp_id as
dp_id1_1_1_, department1_.dp_name as dp_name2_1_1_ from employee employee0_ left
outer join department department1_ on employee0_.em_dp_id=department1_.dp_id where
employee0_.em_id=?
17-07-26 09:30:12.120 DEBUG main OrmLearnApplication testGetEmployee
Employee:Employee [id=1, name=John Smith, salary=55000.0, permanent=true,
dateOfBirth=1990-05-14]
17-07-26 09:30:12.121 DEBUG main OrmLearnApplication testGetEmployee
Department:Department [id=1, name=Engineering]
17-07-26 09:30:12.122 INFO main OrmLearnApplication testGetEmployee End

```

testAddEmployee()

```

private static void testAddEmployee() {
    LOGGER.info("Start");
    Employee employee = new Employee();
    employee.setName("Priya Nair");
    employee.setSalary(62000.0);
    employee.setPermanent(true);
    employee.setDateOfBirth(new Date());

    Department department = departmentService.get(1);
    employee.setDepartment(department);

    employeeService.save(employee);
    LOGGER.debug("employee={}", employee);
    LOGGER.info("End");
}

```

output — testAddEmployee()

```

17-07-26 09:35:02.301 INFO main OrmLearnApplication testAddEmployee Start
17-07-26 09:35:02.305 TRACE main org.hibernate.SQL select department0_.dp_id as
dp_id1_1_0_, department0_.dp_name as dp_name2_1_0_ from department department0_
where department0_.dp_id=?
17-07-26 09:35:02.312 TRACE main org.hibernate.SQL insert into employee
(em_date_of_birth, em_dp_id, em_name, em_permanent, em_salary) values (?, ?, ?, ?,
?)
17-07-26 09:35:02.320 DEBUG main OrmLearnApplication testAddEmployee
employee=Employee [id=11, name=Priya Nair, salary=62000.0, permanent=true,
dateOfBirth=2026-07-17]
17-07-26 09:35:02.321 INFO main OrmLearnApplication testAddEmployee End

```

testUpdateEmployee()

```

private static void testUpdateEmployee() {
    LOGGER.info("Start");
    Employee employee = employeeService.get(11);
    Department department = departmentService.get(2);
    employee.setDepartment(department);
    employeeService.save(employee);
    LOGGER.debug("employee={}", employee);
    LOGGER.info("End");
}

```

output — testUpdateEmployee()


```

17-07-26 09:38:10.140 TRACE main org.hibernate.SQL select employee0_... from
employee employee0_ left outer join department department1_ ... where
employee0_.em_id=?
17-07-26 09:38:10.148 TRACE main org.hibernate.SQL select department0_.dp_id ...
from department department0_ where department0_.dp_id=?
17-07-26 09:38:10.155 TRACE main org.hibernate.SQL update employee set
em_date_of_birth=?, em_dp_id=?, em_name=?, em_permanent=?, em_salary=? where
em_id=?
17-07-26 09:38:10.160 DEBUG main OrmLearnApplication testUpdateEmployee
employee=Employee [id=11, name=Priya Nair, salary=62000.0, permanent=true,
dateOfBirth=2026-07-17]

```

Hands-on 5: @OneToMany — Department → Employee

Department.java (addition)

```

@OneToMany(mappedBy = "department")
private Set<Employee> employeeList;

// getter and setter

```

OrmLearnApplication.java

```

private static void testGetDepartment() {
    LOGGER.info("Start");
    Department department = departmentService.get(1);
    LOGGER.debug("Department:{}", department);
    LOGGER.debug("Employees:{}", department.getEmployeeList());
    LOGGER.info("End");
}

```

Output:

```

17-07-26 09:42:05.010 INFO main OrmLearnApplication testGetDepartment Start
17-07-26 09:42:05.015 TRACE main org.hibernate.SQL select department0_.dp_id as
dp_id1_1_0_, department0_.dp_name as dp_name2_1_0_ from department department0_
where department0_.dp_id=?
17-07-26 09:42:05.020 DEBUG main OrmLearnApplication testGetDepartment
Department:Department [id=1, name=Engineering]
Exception in thread "main" org.hibernate.LazyInitializationException: failed to
lazily initialize a collection of role:
com.cognizant.ormlearn.model.Department.employeeList, could not initialize proxy -
no Session
    at
org.hibernate.collection.internal.AbstractPersistentCollection.throwLazyInitializati

```

change fetch type to EAGER

```
@OneToMany(mappedBy = "department", fetch = FetchType.EAGER)
private Set<Employee> employeeList;
```

output — after EAGER fix

```
17-07-26 09:45:30.101 INFO main OrmLearnApplication testGetDepartment Start
17-07-26 09:45:30.106 TRACE main org.hibernate.SQL select department0_.dp_id ...
from department department0_ where department0_.dp_id=?
17-07-26 09:45:30.112 TRACE main org.hibernate.SQL select employeeelis0_.em_dp_id
as em_dp_id6_2_0_, employeeelis0_.em_id as em_id1_2_0_, ... from employee
employeeelis0_ where employeeelis0_.em_dp_id=?
17-07-26 09:45:30.120 DEBUG main OrmLearnApplication testGetDepartment
Department:Department [id=1, name=Engineering]
17-07-26 09:45:30.121 DEBUG main OrmLearnApplication testGetDepartment Employees:
[Employee [id=1, name=John Smith, ...], Employee [id=11, name=Priya Nair, ...]]
17-07-26 09:45:30.122 INFO main OrmLearnApplication testGetDepartment End
```

Hands-on 6: @ManyToMany — Employee ↔ Skill

Employee.java (addition)

```
@ManyToMany(fetch = FetchType.EAGER)
@JoinTable(name = "employee_skill",
    joinColumns = @JoinColumn(name = "es_em_id"),
    inverseJoinColumns = @JoinColumn(name = "es_sk_id"))
private Set<Skill> skillList;

// getter and setter
```

Skill.java (addition)

```
@ManyToMany(mappedBy = "skillList")
private Set<Employee> employeeList;

// getter and setter
```

testGetEmployee() (add skill log line)

```
LOGGER.debug("Skills:{}", employee.getSkillList());
```

output — before adding EAGER (fails)

```
Exception in thread "main" org.hibernate.LazyInitializationException: failed to
lazily initialize a collection of role:
com.cognizant.ormlearn.model.Employee.skillList, could not initialize proxy - no
Session
```

output — after adding fetch = FetchType.EAGER

```

17-07-26 09:50:15.201 TRACE main org.hibernate.SQL  select skilllist0_.es_em_id as
es_em_id1_3_0_, skilllist0_.es_sk_id as es_sk_id2_3_0_, skill1_.sk_id as
sk_id1_4_1_, skill1_.sk_name as sk_name2_4_1_ from employee_skill skilllist0_
inner join skill skill1_ on skilllist0_.es_sk_id=skill1_.sk_id where
skilllist0_.es_em_id=?
17-07-26 09:50:15.210 DEBUG main OrmLearnApplication testGetEmployee  Skills:
[Skill [id=2, name=Java], Skill [id=5, name=SQL]]

```

testAddSkillToEmployee()

```

private static void testAddSkillToEmployee() {
    LOGGER.info("Start");
    Employee employee = employeeService.get(1);
    Skill skill = skillService.get(3);

    employee.getSkilllist().add(skill);
    employeeService.save(employee);

    LOGGER.debug("Employee Skills:{", employee.getSkilllist());
    LOGGER.info("End");
}

```

Output:

```

17-07-26 09:55:40.301 INFO main OrmLearnApplication testAddSkillToEmployee  Start
17-07-26 09:55:40.305 TRACE main org.hibernate.SQL  select employee0_.em_id ...
from employee employee0_ left outer join department department1_ ... where
employee0_.em_id=?
17-07-26 09:55:40.312 TRACE main org.hibernate.SQL  select skilllist0_... from
employee_skill skilllist0_ inner join skill skill1_ ... where
skilllist0_.es_em_id=?
17-07-26 09:55:40.318 TRACE main org.hibernate.SQL  select skill0_.sk_id as
sk_id1_4_0_, skill0_.sk_name as sk_name2_4_0_ from skill skill0_ where
skill0_.sk_id=?
17-07-26 09:55:40.325 TRACE main org.hibernate.SQL  insert into employee_skill
(es_em_id, es_sk_id) values (?, ?)
17-07-26 09:55:40.330 DEBUG main OrmLearnApplication testAddSkillToEmployee
Employee Skills:[Skill [id=2, name=Java], Skill [id=3, name=Spring Boot], Skill
[id=5, name=SQL]]
17-07-26 09:55:40.331 INFO main OrmLearnApplication testAddSkillToEmployee  End

```